

Andrew Zheng

andrew.zheng1@uwaterloo.ca | [in/andrewzheng2007](https://in.linkedin.com/in/andrewzheng2007) | github.com/awzheng | awzheng.me

EXPERIENCE

Software Engineer Intern | [Miovision](#) | Waterloo, ON

May 2026 – Aug 2026

- Re-engineered RAUC firmware update path for Yocto-based x86-64 MioconnectOS Linux devices in Node.js via adaptive block hash index HTTP streaming through AWS IoT to reduce OTA bandwidth cost by **98%** (1.4GB to 21MB)
- Built end-to-end HIL release pipeline for ARM64 Linux CV devices in Node.js and GitLab CI on Terraform-provisioned, Ansible-configured OTA runners via AWS IoT to poll **15** health metrics over **48h** with **<1%** frame-drop
- Deployed virtual 360° camera emulator app on Firecracker-microVM (Go/Docker); shipped H.264 RTSP + camera-API to ARM64 embedded Linux CTM devices to enable camera-free HIL development and cut hardware costs
- Architected fleet-wide dual lookup for device search engine in Kotlin/Spring Boot to stream schema-registry-backed Avro events over Kafka to Kubernetes-hosted index, eliminating **100%** of stale results across **43K+** devices

Embedded Systems Developer | [Waterloo Rocketry](#) | Waterloo, ON

Sep 2025 – Present

- Designed 4-layer recovery parachute arming PCB for the **world record highest amateur liquid rocket (63,497 ft)**, integrating dual 9V LiPo power stages, INA180 current-sense amplifiers, and 15V/4.1V ADC voltage scaling
- Engineered hardware fail-safe for PIC18 microcontroller to drive a P/N-channel MOSFET network to arm dual altimeters, using passive gate biasing to guarantee parachute deployment during complete MCU Hi-Z faults

Founder & Software Engineer | [Case Tutoring Startup](#) | Toronto, ON

May 2024 – Aug 2025

- Drove **\$10K+** revenue through GTM campaign for self-founded case coaching service, scaling across **10+** schools
- Developed React/Next.js AI study app with adaptive spaced repetition algorithm for **150+** high school students
- Won **1st Place** globally and coached **50+** qualifiers/**30+** finalists at DECA ICDC, America's largest business contest

PROJECTS

Allocate | [macOS CPU Core Governor](#) | *Rust, XNU QoS, Darwin API, Mach IPC, POSIX*

[GitHub](#) 

- Built system-level macOS orchestrator to intelligently thread processes across P/E cores and override default scheduling, using XNU QoS and POSIX thread policies to cut background CPU consumption by **79%** with **100%** app uptime
- Developed Rust FFI daemon to monitor NSWorkspace lifecycle events, sample Darwin proc_pidinfo and Mach CPU ticks per 500ms, using raw libxpc configuration and override IPC to reroute processes to appropriate CPU cores
- Released end-to-end SwiftUI/AppKit/Charts macOS app with 60-sample live CPU chart, compute telemetry, battery monitoring, configurable CPU thresholds, and two-way Mach/XPC IPC for real-time, app-specific P/E core control

GitIRL Bot | [Hack the North 2026 Winner](#) | *Python, FastAPI, SLAM, OpenAI API*

[Devpost](#) 

- Won Best use of Bracket Bot hardware by building an autonomous world-state version control system to map physical objects to versioned snapshots, detect environmental changes, and command a humanoid robot to restore prior states
- Designed distributed orchestration/perception pipeline across Bracket Bot and cloud services, indexing **6.9M+** telemetry points in Elasticsearch to enable hybrid semantic search and translate natural-language input into robotic actions

Albedo | [macOS Fleet Thermal Control Agent](#) | *Rust, AppleSMC, IOKit, Prometheus, Grafana*

[GitHub](#) 

- Built Apple Silicon fleet thermal controller to send metrics and PID-level E-core override commands to Allocate via launchd XPC Mach service; achieved **12°C/22°F** CPU cooling with **100%** app uptime without hardware fans
- Engineered 18-sensor Rust thermostat that samples AppleSMC and IOReport to translate CPU/GPU/ANE telemetry, SoC energy, DVFS residency, and per-process CPU time into thermal-throttling, clock-loss, and power usage insights
- Shipped end-to-end SwiftUI menu-bar app + Prometheus/Grafana fleet observability pipeline to derive telemetry, record before and after thermal effects, and remotely write device-labeled metrics for throttling and performance analysis

CrawlStars | [Distributed Search Engine](#) | *Go, JavaScript, MongoDB, REST API*

[GitHub](#) 

- Engineered concurrent crawlers in Golang producer-consumer architecture to process **2000+** pages/min in **<15MB**
- Optimized sync.Map deduplication to enable O(1) lookup for **50K+** URLs and filter **70%** of redundant crawls
- Deployed web search engine to calculate SEO relevance via MongoDB Atlas aggregation pipeline for **50K+** URLs

SKILLS

Languages: C, C++, Python, Rust, Go, Java, Kotlin, Swift, JavaScript, TypeScript, HTML, CSS, Ruby, Verilog, Bash

Infrastructure: AWS, Linux, Docker, Kubernetes, Terraform, Ansible, Kafka, CI/CD, Git, Prometheus, Grafana

Backend: Node.js, Express.js, FastAPI, Spring Boot, Ruby on Rails, REST API, MongoDB, Jupyter

Frontend/Other: React, Next.js, Tailwind, Streamlit, XGBoost, SLAM, VLA, Ubuntu, Mach, Apple IOKit

EDUCATION

University of Waterloo | *BASc, Computer Engineering*

Sep 2025 – Apr 2030

Allocate

Xcode

100% (Battery)

Live

#	Process	CPU	RAM	Threads
1	Xcode	0.2%	257 MB	15
2	java	7.3%	2.8 GB	176
3	mobileassetd	0.8%	40 MB	8
4	coreaudiod	0.2%	46 MB	12
5	sysmond	0.2%	3 MB	3
6	WindowServer	0.1%	309 MB	23
7	Activity Monitor	0.1%	68 MB	6

Designed XPC to control Allocate!

Albedo

0%

47.2 °C

50.6%

0.179

30.1%

CPU 43° 0.7 W

GPU 46° 0.0 W

ANE 33° 0.0 W

Left fan 2319 RPM

Right fan 2514 RPM

76% Clock 20% Load 0.23 Perf/W

Top energy users

Discord Helper (Ren...

Thermostat

Hold 85°C

release 73°C

Allocate connected — can move apps to E-cores

Nothing to cool

Grafana Individual edge device view

Mac User

View CPU usage; set rules

Allocate app SwiftUI + Swift Charts

App switch event

AppKit / NSWorkspace Foreground-app events

macOS Observability libproc + IOKit

CPU, memory, threads, battery

System snapshot

Schedule work across cores

Throttles, pause, manual overrides

Process table + CPU chart

Protect active app

Configuration

Mach IPC / XPC

Live telemetry

Allocate Core Daemon + Governor

Throttle heavy background work

taskpolicy Apple scheduling control

Apply background / restore policy

XNU QoS Policy

Prioritize foreground

Apple Silicon Scheduler Performance + Efficiency Cores

Mac User

View and control

SwiftUI Menu Bar

C ABI snapshots

Inspect and govern

Ratatui CLI

Native Rust calls

Rust Thermal Engine

Expose derived metrics

Metrics HTTP API

Request E-core hold

Allocate XPC Daemon

Apply QoS policy

Apple Silicon Scheduler

Scrape host metrics

Grafana Alloy

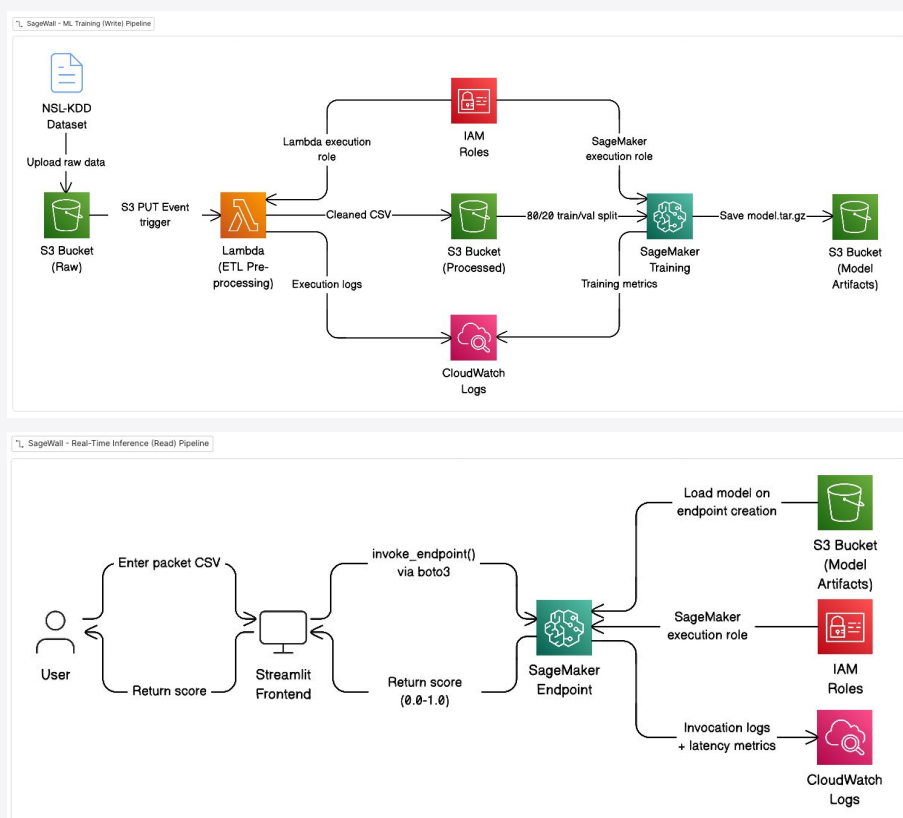
Remote write

Prometheus TSDB

Fleet visualization

Grafana Dashboards

Grafana fleet control host view



Golang, MongoDB | O(1) lookup time | 2000+ URLs/minute | <15MB memory usage

